PROJECT PROGRESS REPORT

(Progress Report no. 1)

1. Student Information

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Enrollment Number	25IC03CA053
Program/Branch	Master of Computer Applications
Semester	3
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2. Project Details

Type	Research Project
Department/Domain	MCA CA(DS/ML) / AI & ML
Mentor Name	Srikant Singh
Start Date	08-06-2026
End Date	03-10-2026

3. Work Progress Summary

a) Project Title:

OpenCV - Based System for Crack Identification and Width Estimation

b) Problem Statement:

Building structures often develop cracks due to aging, environmental conditions, heavy loads, and poor construction quality. Manual inspection is time-consuming, expensive, and prone to human error. This project develops an OpenCV - based system to automatically detect building cracks from images and accurately estimate crack width, enabling faster inspection, timely maintenance, and improved structural safety.

c) Objectives:

- To detect cracks in building surfaces using OpenCV image processing.
- To measure crack width accurately from captured images.
- To reduce manual inspection time and improve detection accuracy.
- To provide a cost-effective solution for structural health monitoring.

d) Proposed Methodology:

- Collect images of building cracks
- Preprocess images using grayscale conversion and noise removal
- Detect crack regions using OpenCV image processing techniques.
- Apply edge detection and contour analysis to identify cracks.
- Estimate crack width using pixel measurements and calibration.
- Display the detected crack and measured width through a user-friendly interface.

e) Expected Outcomes:

- Accurate detection of building cracks.
- Precise crack width estimation.
- Reduced inspection time and human effort.
- Improved safety through early crack identification.

f) Future Scope:

- Integrate deep learning (CNN/YOLO) for higher detection accuracy.
- Support real-time crack detection using drones or CCTV cameras.
- Develop a mobile application for on-site inspection.
- Store inspection reports on the cloud for long-term monitoring and analysis.

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4. Tasks Accomplished

- Conducted a literature survey on building crack detection and structural health monitoring.
- Collected and organized a dataset of building crack images from public sources.
- Installed and configured the Python environment with OpenCV, NumPy, and Matplotlib.
- Studied OpenCV image processing techniques such as grayscale conversion, filtering, thresholding, edge detection, and contour detection.
- Designed the project workflow for crack detection and width estimation.
- Implemented the initial crack detection algorithm using OpenCV.
- Performed preliminary testing on sample images and analyzed the detection results.
- Prepared the project proposal, problem statement, objectives, and methodology documentation.

5. Tasks in Progress / Next Plan

- Create a LOGO And UI/UX Planning and Design Create in Figma.
- Develop the Admin Module (Login, Dashboard, User Management, Dataset Management, View Reports, System Settings).
- Develop the User Module (Registration, Login, Image Upload, Crack Detection, View Results, Download Report, History).
- Implement Node.js & Express.js REST APIs and connect them with MongoDB for data storage and retrieval.
- Integrate the Python OpenCV crack detection and width estimation module with the MERN application.
- Perform testing, fix bugs, optimize performance, generate reports, and prepare the final project documentation and presentation.
- Complete the implementation of the crack detection algorithm using OpenCV.
- Improve crack width estimation accuracy through image calibration and measurement techniques.
- Develop a graphical user interface (GUI) for image upload and result visualization.
- Test the system using different building crack datasets under various lighting conditions.

- Evaluate system performance based on accuracy, precision, and processing time.
- Prepare the final project report, presentation, and demonstration with experimental results.

6. Learnings / Skills Acquired

- Learned Python programming for image processing and computer vision applications.
- Gained practical experience with OpenCV for image preprocessing, crack detection, contour extraction, edge detection, and crack width estimation.
- Developed responsive React.js interfaces for Admin and User dashboards with image upload and result visualization.
- Gained experience in MongoDB for managing users, crack images, detection results, and reports.
- Learned Python and OpenCV for image preprocessing, crack detection, and crack width estimation.
- Improved skills in authentication, CRUD operations, API integration, debugging, testing, and project documentation.
- Gained experience in integrating OpenCV-based Python modules with a web application.
- Enhanced debugging, testing, version control, project documentation, and software deployment skills.
- Improved teamwork, problem-solving, software design, and project management skills through practical implementation.

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Evaluation (To be filled by Mentor)

Criteria	Bloom's Level	Excellent (20)	Good (15)	Average (10)	Poor (5)	Marks Obtained (Out of 20)
Problem Definition	L4–Analyze	Clearly defined, justified real-world problem	Defined but lacks depth	Vague problem	No clear problem	
Project Title Relevance	L2–Understand	Highly precise and aligned	Mostly relevant	Partially relevant	Misleading	
Scope of Work	L5–Evaluate	Well-defined scope, deliverables, constraints	Defined with minor gaps	Broad/unclear	No scope	
Related Study	L4–Analyze	3–5 sources analyzed critically	Sources listed	Minimal references	No study	
Clarity & Presentation	L3–Apply	Structured, professional	Minor issues	Some confusion	Poor	
Total Marks						

Evaluation

Student Signature	Mentor (Institute) Signature

Mentor (Institute) Remarks

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